

1. (2 points) How many atmospheres of pressure does a squid living 560 meters deep in the Pacific Ocean experience?

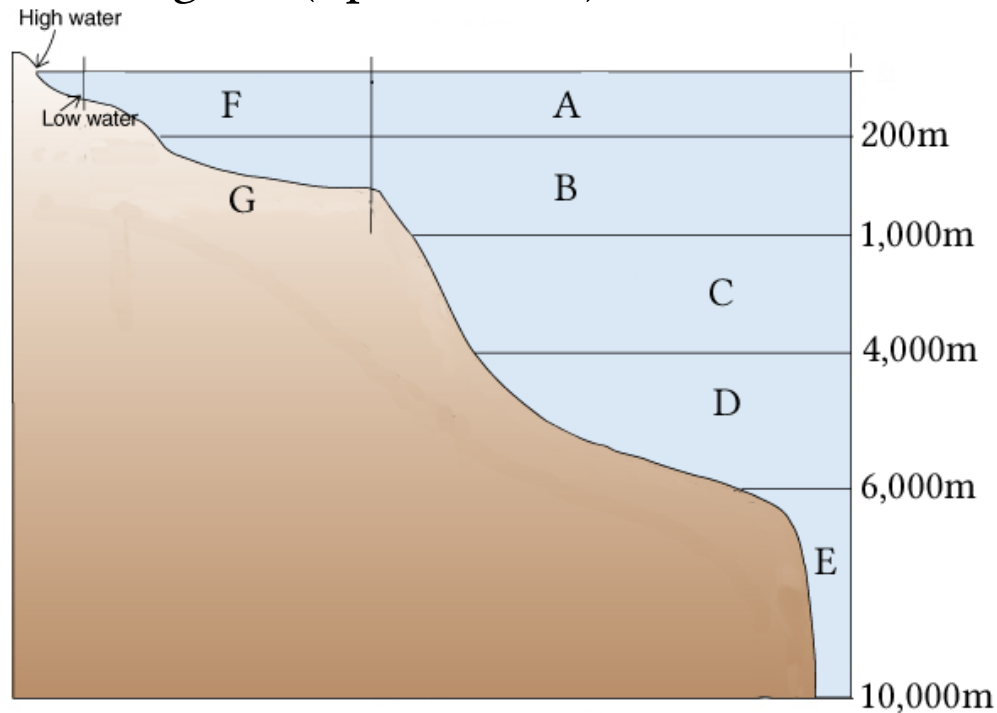
2. (2 points) Chloride and sodium are the two most common ions in sea water. List the next four most common ions in sea water, from highest to lowest concentration. Do not list the concentration of each ion. $\frac{1}{2}$ point per ion.

3. (3 points) List the three deepest ocean trenches and the ocean in which each trench is located. $\frac{1}{2}$ point per blank.

Trench Name

Ocean Basin

Use correct names to fill in the blanks below for the zones identified by letters in the figure. (1 point each.)



6. _____ A, based on depth.
7. _____ B, based on light.
8. _____ C, based on depth.
9. _____ C, D, and E combined, based on light.
10. _____ F, open water out to shelf break.
11. _____ G, benthic, from low water mark.

Extra credit: Proper names for D and E. ($\frac{1}{2}$ point each.)

D. _____ E. _____

12. (10 points) See of figure of Africa in separate handout.

The gladiator crab *Callinectes gladiator* lives in estuaries along the west coast of Africa (indicated by the dots). The crab larvae develop over the continental shelf. In spring, the local winds blow from north to south. In fall, the winds blow from south to north. State which season a crab living in Bengo Bay should release its larvae to ensure the larvae reach the shelf and then return to the estuary for metamorphosis to adults. Explain why, using the information provided. Include the two forces involved and describe why or how they help steer the larvae in the local surface currents.

13. (a) (5 points) On screen is the Indo-Pacific Snaggletooth (*Astronesthes indopacificus*). Tell whether the snaggletooth more likely lives in the mesopelagic or bathypelagic zone. Explain one adaptation, other than bioluminescence and *visible in the image*, the fish exhibits to justify your answer. The adaptation *must be characteristic of the depth zone* and not common to all deep-sea fishes. Be sure to explain the specific relationship between the adaptation and the zone you choose.

- (b) (5 points) (Snaggletooth, con't.) Identify a different adaptation, other than bioluminescence and *visible in the image*, that is specific to life in the deep-sea (this adaptation can be common to deep sea organisms regardless of depth zone). Explain how that adaptation is related to the environment or survival in the deep-sea.

14. (a) (15 points) Summarize the classic model of primary productivity in the North Atlantic Ocean from late winter to early summer. Your summary should name the major phytoplankton and zooplankton organisms and describe their roles in overall primary production. Your summary should also describe the seasonal availability of nutrients and light to explain the timing of peak productivity. You may include an illustration. Do not discuss the fall season.

- (b) (5 points) Name and describe the modification to the classic model to account for the amount of primary productivity and retention of POM and DOM in tropical surface waters.

Extra credit: For one point, tell me what would you *see* if you walked along the abyssal plain from Maine to Spain. Why? This is a trick question; I do have a sense of humor.

15. (10 points) Describe the difference between active and passive margins. Describe how this affects the biology of benthic (bottom-dwelling) organisms living on the different margin types. Be clear in your example.